

Machine Learning Engineer Home Assignment

Fake vs Real News Classification (NLP + GenAI)

Thank you for considering joining us at AMEX GBT! We're excited about the possibility of working together to make a meaningful impact in the travel industry. Your expertise and creativity could play a key role in shaping the future of travel experience for businesses and their employees around the globe.

The objective of this assignment is to evaluate your approach, methodology, and problem-solving skills in handling an AI task. The focus is on understanding your reasoning and thought process rather than obtaining a "perfect" solution.

Task Instructions

You are tasked with exploring and leveraging a provided text dataset consisting of news articles labelled as **Fake** or **Real** across two CSV files.

Your goal is to propose and implement a solution that can classify an unseen article as **Fake** or **Real**, while balancing model quality with practical constraints.

We recommend coding locally in **Anaconda Jupyter Notebook**, using **Google Colab**, or any other environment you're comfortable with. Feel free to choose the setup that works best for you, as long as you ensure your solution is **reproducible** when shared.

Business framing / constraints

Assume the model may be used in an online setting. Your solution should explicitly consider:

- Latency / cost trade-offs for inference
- Operational simplicity (how easy it is to deploy and maintain)
- Reliability under messy or slightly altered input text
- Reproducibility (another engineer should be able to run your work end-to-end)

Submission Guidelines (Requirements)

1) Deliverables

Submit a single zip archive containing a **Python-based project** of your choice (e.g., Python scripts, a notebook, a small package, or a combination), as long as it can be run end-to-end and reproduces your results.

Your submission must include:

A. Data preparation

- Load both CSVs and create a unified dataset with a clear target label.
- Document your preprocessing choices.
- Describe key pitfalls you considered (e.g., leakage risks, duplicates, split strategy).

B. EDA

- Brief dataset summary (size, columns, missing values, class balance).
- A short set of observations about the text and labels.
- At least **2** visualizations saved to disk.

C. Modeling (two approaches required)

Implement and compare **two distinct approaches**:

Track 1 — Production-oriented baseline

- A strong baseline you believe is realistic for **fast, low-cost online inference** and easy deployment/maintenance.

Track 2 — GenAI-based approach

- A solution that leverages **a generative LLM** in the loop (e.g., classification and/or LLM-assisted workflow).
- You may use an external API and/or a locally runnable model.

For both tracks, briefly justify *why* the approach is appropriate given the business framing (latency/cost/complexity).

D. Evaluation (required)

Report at minimum:

- Accuracy
- Precision / Recall / F1 (per class + overall)
- Confusion matrix

You may add any additional metrics or analyses you consider relevant (e.g., calibration, latency/cost estimates, robustness metrics) and briefly justify how they influence the comparison between Track 1 and Track 2.

E. Robustness (required, lightweight)

Demonstrate at least **two** robustness checks that reflect real usage (e.g., partial text, noisy text, reordered content, irrelevant additions) and summarize the observed impact.

F. Insights & limitations (required)

- Key learnings from the data and model behavior
- Typical failure cases and what they suggest
- Concrete improvement ideas (data, modeling, evaluation, monitoring)
- If using GenAI: brief note on privacy/cost/compute considerations

Required project components

- A dependency specification (requirements.txt, pyproject.toml, or equivalent)
- Saved outputs needed to reproduce results (plots/metrics; model artifacts optional)

2) Handling secrets (API keys)

If your solution uses hosted services (e.g., an LLM API):

- **Do not include any secrets** (API keys/tokens) in the submitted zip.
- It is acceptable if the GenAI portion requires a key available only on your machine (e.g., provided via environment variable) and can be demonstrated during the interview screen-share.
- Please include brief instructions on how to enable/disable the GenAI component (e.g., “set OPENAI_API_KEY” or “run with --no-llm”), so the rest of the project still runs without secrets.

Notes

- You have **1 week** to submit your solution.
- The assignment is open-ended; there is no single correct solution. We encourage creativity and critical thinking.
- You will be expected to share your screen to re-run and potentially modify your solution during the technical interview.

Best regards,

Product AI Engineering team @ Amex GBT